

Key: A Country Nobody Lives In

84-355 Democracy's Data | Demographics | Instructor copy — do not distribute

Jonathan Cervas

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All code is given to students. Grade the interpretation.

Structure. The brief (`demographics-brief.Rmd`) is a chapter, “A Country Nobody Lives In,” organised under prose headings and carrying eight numbered figures: Figure 1, the eight categories in true proportion with the small tail magnified; Figure 2, which box you could be counted in by census year, with the two OMB revisions drawn as rules; Figure 3, the six races counted alone and alone-or-in-combination; Figure 4, 2010 against 2020 with the processing discontinuity shaded; Figure 5, six small-multiple maps of the states; Figure 6, the diversity staircase with its random-shuffle null; Figure 7, Wayne County at three scales; Figure 8, the blocks along Eight Mile Road and the composition profile across it. The lab (`demographics.Rmd`) runs Steps 1–11 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab.** The chapter covers the same ground in the same order, so each lab Step has a section facing it: Step 1 = “What the census counted” (Figure 1); Step 2 = “Nobody at the Census Bureau chose these boxes” (Figure 2); Step 3 = the prediction prompt and Figure 3; Step 4 = “A decade that mostly did not happen” (Figure 4); Steps 5 and 6 = “Where people are” (Figure 5); Steps 7 and 8 = “The number that keeps moving” (Figure 6); Step 9 = “One county with no majority” (Figure 7); Step 10 = “One street” (Figure 8); Step 11 = “What this data can and cannot support.”

What this lab is for, and what it deliberately is not. Three labs in this course already own parts of this territory and this one stays out of all three. `census` owns *how the census defines race* — Questions 8 and 9, the write-in boxes, “Some Other Race.” `areal-units` owns *the modifiable areal unit problem*. `census-geography` owns *the nesting spine*. This lab is the substantive one: what the country actually looks like, and how differently it looks at different scales. Where it touches those labs it says so and points at them; it does not re-teach them.

Every number below is computed from the built files at knit time.

The session in one sentence

The United States scores 60.4 on this chapter’s diversity scale and the average American’s census tract scores 43.1 — and neither number means anything until you know which of the Census Bureau’s two correct counts of race it was built from, and which revision of an OMB directive created the categories it counts.

The findings

```
c(us_population      = FN("us_pop"),
  diversity_nation    = FN("div_nation"),
  diversity_state     = FN("div_state"),
  diversity_county    = FN("div_county"),
  diversity_tract     = FN("div_censustract"),
  diversity_block     = FN("div_censusblock"),
  H_tract            = FN("H_censustract"),
  H_tract_null       = FN("H0_censustract"),
  pct_in_tract_80plus = FN("pop_tract_80plus"))
```

```
#>      us_population      diversity_nation      diversity_state      diversity_county
#>      331449281.0000           60.4000           55.0000           50.4000
#>      diversity_tract      diversity_block           H_tract           H_tract_null
#>           43.1000           36.2000           0.2866           0.0007
#> pct_in_tract_80plus
#>           33.9000
```

```
c(white_alone        = FN("us_white_alone"),
  white_any           = FN("us_white_any"),
  aian_alone          = FN("us_aian_alone"),
  aian_any            = FN("us_aian_any"),
  aian_ratio          = FN("us_aian_ratio"),
  white_ratio         = FN("us_white_ratio"))
```

```
#>  white_alone  white_any  aian_alone  aian_any  aian_ratio  white_ratio
#> 204277273.00 235411507.00 3727135.00 9666058.00         2.59         1.15
```

```
c(chg_white_alone_2010_2020 = FV("chg_white_alone"),
  chg_white_any_2010_2020   = FV("chg_white_any"),
  chg_two_plus              = FV("chg_two_plus"),
  chg_sor_any               = FV("chg_sor_any"))
```

```
#> chg_white_alone_2010_2020  chg_white_any_2010_2020      chg_two_plus
#>           "-8.6"           "+1.9"           "+275.7"
#>           chg_sor_any
#>           "+129.5"
```

- **The White population both fell and rose over the decade.** Alone: -8.6%. Alone or in any combination: +1.9%. Same censuses, same forms, same answers. This is the chapter's cold open and the single best thing in the session.
- **American Indian and Alaska Native goes from 3,727,135 to 9,666,058 — a factor of 2.59.** White moves by 31,134,234 people and a factor of only 1.15. Largest absolute mover and largest proportional mover are different categories; both answers to “which is most affected” are right.
- **The staircase: 60.4 → 55.0 → 50.4 → 43.1.** The random-shuffle null stays at 60.4 through the tract rung, so this is sorting, not small-unit arithmetic.
- **33.9% of Americans live in a tract that is at least 80% one category;** 13.1% in one at least 90%.
- **Wayne County, Michigan is in the most diverse 6.2% of 3,143 counties (rank 196), and 51.8% of its residents live in a tract that is at least 80% one group.** Both sentences are true.
- **Across Eight Mile Road the Black share falls 37.2 points — 90.7% in the last 500 m inside Detroit, 53.5% in the first 500 m outside.** Flat, step, flat.

Wayne County, the inventory

```
table(largest_group_in_tract = wtr$largest)
```

```
#> largest_group_in_tract
#>           AIAN           Asian           Black           Hispanic           NHPI
#>           1           6           279           21           3
#> Some Other Race   Two or More           White
#>           5           5           307
```

```
c(tracts = nrow(wtr),
  tracts_with_people = FN("wayne_tracts_pop"),
  median_pct_one_group = FN("wayne_tract_median_largest"),
  at_least_80_black = FN("wayne_tracts_80_black"),
  at_least_80_white = FN("wayne_tracts_80_white"),
  people_in_those_pct = FN("wayne_pop_80_pct"),
  at_least_90_any = FN("wayne_tracts_90"),
  people_in_those_pct = FN("wayne_pop_90_pct"))
```

```
#>           tracts   tracts_with_people median_pct_one_group
#>           627.0           600.0           81.6
#>   at_least_80_black   at_least_80_white   people_in_those_pct
#>           201.0           125.0           51.8
#>   at_least_90_any   people_in_those_pct
#>           156.0           22.9
```

307 tracts where the largest group is White, 279 where it is Black, and almost nothing that is genuinely mixed. That near-tie at the county level is exactly why the county looks integrated and exactly why it is not. Say the two numbers out loud together; the room usually hears the tie before it hears the split.

The three things to get said

1. Every count of race has at least two correct values, and the reason is dated. Step 3 is where this lands. Make somebody say *why* it is worse than either number alone: if the two counts differed only occasionally you would call it rounding, but the gap is systematically largest for the smallest categories — 2.59× for American Indian and Alaska Native, 1.15× for White — which means the choice of definition moves the groups that civil-rights enforcement is written about, and barely touches the group it is not. Then give the date: the ambiguity did not exist before the 1997 revision of OMB Directive 15, because the form allowed one mark.

2. A decade of demographic change that mostly was not one. Step 4. The Bureau’s own analysts attribute the jump in Some Other Race “largely” to the improved form, processing and coding. Push the class past “so the data is unreliable,” which is the failure mode, and onto the harder point: **2010 and 2020 were the easy pair.** Same standard, same categories, same file layout, same field order. Everything that could be held constant was, and the comparison still fails. Ask what that implies for any series crossing 1997, or any series that will cross the 2024 revision.

3. Step 8 is the session. Two lines on one chart. The observed staircase falls from 60.4 to 43.1; the random-shuffle null does not move. **The lesson is not that America is segregated** — students arrive believing that. The lesson is that a national statistic and a neighbourhood statistic are answers to different questions, and that the gap between them is measurable, is 0.2866 on Theil’s index, and survives the obvious objection.

The line for the board: the country is not a place anybody lives.

Walking the steps

Step	What it does	Min	Watch for
1 The eight categories	331,449,281 people, 8 boxes, no remainder	4	Somebody will ask why Hispanic is not a race. That is census ’s lab; give the one-sentence answer and move on.
2 Who chose the boxes	Three issues of OMB Directive 15	8	Do not skip. The “Two or More Races” row being empty before 2000 is the whole point of the figure.
3 Alone or in combination	2.59× against 1.15×	12	The prediction. Take the vote before revealing.
4 Two censuses	-8.6% and +1.9%	10	The trap is “so the data is unreliable.” Push to “the easy pair still failed.”
5 Regions	West 65.0, Midwest 47.0	4	Regions are too big to be interesting. Say so and keep moving.
6 States	Hawaii 75.6, Maine 22.8	4	Only 4 states have a largest group that is not White.
7 The measure	Why entropy, not dissimilarity	8	The null is introduced here, not in Step 8. Make sure they know what it controls for before they see it.
8 The ladder	60.4 → 43.1	12	The pivot. Ask what the dashed line would look like if the fall were an artefact.
9 One county	Wayne: no majority, and 51.8% in a lopsided tract	10	Both facts, in the same breath. The tie between 307 and 279 tracts is the moment.
10 One street	37.2 points across one road	5	The <i>shape</i> — flat, step, flat — not the levels.

Step	What it does	Min	Watch for
11 What you can say	Including what you cannot	3	Make them say the definitional caveat unprompted.

Total **80 minutes**.

Step 3 deserves the prediction, and it is the best one in the course. Ask the room: *when you let people mark more than one race, which category grows most in proportion?* Almost everyone says Black or White, because those are the two categories the first two steps put in their heads. It is **American Indian and Alaska Native**, which nearly triples, and the mechanism — the smaller the group, the larger the share of its identifying population that is also something else — is obvious in hindsight and essentially never guessed.

If you are short on time, protect 3, 4 and 8. The two-definitions reveal, the decade that did not happen, and the staircase with its null are the argument. Steps 5 and 6 can be read aloud from the tables in ninety seconds.

Two things that will come up, and the honest answers

“Isn’t this just the modifiable areal unit problem again?” Yes and no, and the distinction is worth eight minutes because a sharp student will raise it. **areal-units** shows that the *statistic* changes with the unit — variance collapses, extremes vanish, a correlation flips sign — and concludes that the unit is a choice with consequences. That is a fact about the instrument. Step 8 runs the same shape of ladder and then **subtracts the instrument**: shuffle the same people at random into units of the identical sizes and the staircase disappears.

```
lad[, c("level", "units", "diversity", "H", "H0")]
```

```
#>      level  units diversity      H      H0
#> 1 Nation      1  60.40302 0.00000000 0.00004172145
#> 2 Region      4  58.29831 0.03484442 0.00002913835
#> 3 Division     9  57.27177 0.05183928 -0.00003226237
#> 4 State     51  55.02968 0.08895805 -0.00005777551
#> 5 County   3143  50.35530 0.16634464 0.00001706186
#> 6 Census tract 84414 43.09214 0.28658969 0.00067687878
#> 7 Block group 239780 42.06346 0.30361987 0.00188103151
#> 8 Census block 8132968 36.22271 0.40031620 0.04067666177
```

Look at H0. It is essentially zero at every rung down to block groups (0.0019), and only at the block rung does it reach 0.0407, against an observed 0.4003. **So roughly 89.8% of even the block-level figure is real.** MAUP is a genuine problem and this is what it looks like when you measure how much of it you have.

“Isn’t the block level ruined by differential privacy?” Partly, and the answer should be precise rather than hand-waving. The 2020 disclosure-avoidance system held **state** total populations invariant and did not alter them; it did **not** hold block totals invariant. Block counts carry the most noise of any level, and this is a second reason — beyond small-unit arithmetic — to treat the last rung of Figure 6 as indicative rather than exact. The tract rung is the one to quote. If a student notices that the null in Step 8 does not separate DP noise from small-N noise, that is a genuinely sophisticated observation and should be rewarded: it does not, and the two are confounded at the block level.

Option A — A different county

Scope: all 3,143 counties, in `data/counties.csv`.

Verified output

```
big <- cou[cou$total >= 250000, ]
head(big[order(-big$div), c("name", "st", "total", "div", "largest")], 6)
```

```
#>      name st  total  div largest
#> 1870  Queens 36 2405464 75.5 Hispanic
#> 235   Solano 06  453491 74.9   White
#> 549  Honolulu 15 1016508 73.5   Asian
#> 188   Alameda 06 1682353 73.3   Asian
#> 221  Sacramento 06 1585055 72.9   White
#> 2603 Fort Bend 48  822779 72.5   White
```

```
head(big[order(big$div), c("name", "st", "total", "div", "largest")], 5)
```

```
#>      name st  total  div largest
#> 2764   Webb 48 267114 11.4 Hispanic
#> 2632  Hidalgo 48 870781 17.0 Hispanic
#> 2555   Cameron 48 421017 19.7 Hispanic
#> 2310 Westmoreland 42 354663 20.5   White
#> 1773  Rockingham 33 314176 22.3   White
```

3/4: Picks a high-scoring county and a low-scoring one, reports both scores, and says the county score does not tell you whether people live together.

4/4: Names what would settle it — the tract-level distribution inside the county, or the same staircase computed within it — and notices that the lab already did this for Wayne and that Wayne’s county score (57.2) is high.

4/4, best: Notices the **Webb County trap**, which is the reason this option exists. The least diverse large county in the United States is Webb County, Texas, at 11.4 — and its largest group is hispanic, not White. **A low diversity score does not mean “white.”** It means “one group,” and the entropy index is deliberately blind to which. A student who spots that has understood what the measure does and does not encode, and has independently rediscovered why the chapter refused to use a two-group dissimilarity index.

Watch for: a student reading the diversity score as a moral scale, so that Webb County comes out “worse” than a county that is 60% White. Correct it on the spot: the index measures evenness across eight administrative categories and nothing else.

Option B — A different definition

Scope: the state table from Step 6, recomputed with `black_any` in place of `nh_black`.

Verified output

```
sta$pb_nh <- round(100 * sta$nh_black / sta$total, 2)
sta$pb_any <- round(100 * sta$black_any / sta$total, 2)
sta$gap <- round(sta$pb_any - sta$pb_nh, 2)
sta$ratio <- round(sta$black_any / sta$nh_black, 2)

head(sta[order(-sta$gap), c("name", "pb_nh", "pb_any", "gap", "ratio")], 5)
```

```
#>           name pb_nh pb_any gap ratio
#> 40      Rhode Island  5.05  9.11 4.06  1.80
#> 33           New York 13.66 17.49 3.83  1.28
#> 9  District of Columbia 40.91 44.17 3.26  1.08
#> 8           Delaware 21.51 24.74 3.23  1.15
#> 22      Massachusetts  6.50  9.53 3.03  1.47
```

```
head(sta[order(-sta$ratio), c("name", "pb_nh", "pb_any", "gap", "ratio")], 5)
```

```
#>           name pb_nh pb_any gap ratio
#> 27      Montana  0.47  1.17 0.70  2.51
#> 12       Hawaii  1.50  3.21 1.71  2.14
#> 51      Wyoming  0.82  1.67 0.85  2.04
#> 13       Idaho   0.80  1.50 0.70  1.87
#> 40 Rhode Island  5.05  9.11 4.06  1.80
```

The two orderings have almost nothing in common, and that is the finding. By percentage points the biggest mover is Rhode Island (4.06 points). By ratio it is Montana (2.51×), where the Black share is small enough that a modest number of second marks nearly triples it.

3/4: Recomputes, reports that the numbers rise everywhere, and names a few states that move.

4/4: Distinguishes the two orderings and explains why they differ — absolute change tracks where Black populations are large, proportional change tracks where they are small. This is the same structure as Step 3's national result, applied one level down, and saying so is the transfer worth rewarding.

4/4, best: Takes a position on the legal question and defends it. Voting rights litigation under Section 2 generally uses **any-part Black**, and there is a real argument for it: the question is who could form a politically cohesive group, not who ticked exactly one box. There is also a real argument against, since the any-part definition double-counts people across groups and the shares no longer sum to 100. A student who notices **that the any-part columns do not sum to the population** — and that this is a feature of the definition rather than a bug in the file — has read the table properly.

Watch for: adding `black_any` and `nh_white` together and being surprised that they exceed the total. Also watch for a student comparing `black_any` against `nh_black`, which mixes the multiple-race axis with the ethnicity axis; the clean comparison is `black_alone` against `black_any`, and `nh_black` is a third thing again. That confusion is worth ten minutes if it appears, because it is precisely the seam the `census` lab is about meeting the seam this lab is about.

Option C — A different threshold

Scope: `data/tract_largest.csv`, the whole distribution.

Verified output

```
o <- tld
o$pct <- round(100 * o$people / sum(o$people), 1)
o$at_or_above <- round(100 * rev(cumsum(rev(o$people))) / sum(o$people), 1)
o
```

```
#>      bin  people  pct at_or_above
#> 1 [10,20]    2716  0.0      100.0
#> 2 (20,30]  1005701  0.3      100.0
#> 3 (30,40]  15621168  4.7       99.7
#> 4 (40,50]  39890773 12.0       95.0
#> 5 (50,60]  48953005 14.8       82.9
#> 6 (60,70]  52448389 15.8       68.2
#> 7 (70,80]  61051618 18.4       52.4
#> 8 (80,90]  68951967 20.8       33.9
#> 9 (90,100] 43523944 13.1       13.1
```

Reading down the last column: **95.0% of Americans live in a tract whose largest group is over 40%, 52.4% over 70%, 33.9% over 80%, and 13.1% over 90%.**

3/4: Reports the distribution and picks a threshold.

4/4: Notices that the answer can be made to sound like almost anything — “a third of Americans live in a near-segregated neighbourhood” and “two thirds of Americans live in a tract with no dominant group” are both defensible from this one column — and says which they would use and why.

4/4, best: Sees that this is why the chapter leads with an index rather than a threshold. The entropy score has no threshold in it; it is a single number per unit with no cut point to shop for. Then notices the cost of that choice: the index is harder to explain, and “33.9% of Americans” is the sentence that will actually be quoted. **The best answers treat the trade-off between a defensible measure and a repeatable sentence as the real problem, not as a rhetorical flourish.**

Option D — The 2024 categories

Conceptual; no verified output. Grade the reasoning.

```
std[, c("key", "issued", "citation", "categories", "question_form")]
```

```
#>      key      issued      citation categories
#> 1 1977 1977-05-12 Statistical Policy Directive No. 15      5
#> 2 1997 1997-10-30      62 FR 58782      6
#> 3 2024 2024-03-29      89 FR 22182      7
#>      question_form
#> 1 separate collection preferred, not required
#> 2      two separate questions preferred
#> 3      a single combined question required
```



```
cts[cts$c2030 == 1 | cts$c2020 == 1, c("category", "dimension", "c2020", "c2030")]
```

```
#>
#> 1          category dimension c2020 c2030
#> 2          White      race      1      1
#> 3 Black or African American      race      1      1
#> 3 American Indian or Alaska Native      race      1      1
#> 5          Asian      race      1      1
#> 6 Native Hawaiian or Other Pacific Islander      race      1      0
#> 7 Native Hawaiian or Pacific Islander      race      0      1
#> 8      Some Other Race      race      1      0
#> 9      Two or More Races      race      1      1
#> 10 Middle Eastern or North African      race      0      1
#> 11      Hispanic or Latino ethnicity      1      0
#> 12      Hispanic or Latino combined      0      1
```

3/4: Lists what changes — a combined question, 7 categories, MENA added — and says the numbers would be different.

4/4: Sorts the lab’s numbers into three piles correctly. **Unchanged:** the total population; the ladder’s *structure*, since it depends on geography rather than categories. **Changed and predictable in direction:** the diversity score, because it is computed over the number of categories — more categories mechanically raise the ceiling, so 2.0794 becomes $\ln 7$ and every score is rescaled. **Unpredictable:** everything about White and MENA, because nobody knows how people who were counted as White in 2020 will answer a form that offers them a different box.

4/4, best: Sees that the last pile is the interesting one and states the consequence: **a change in the count of the White population between 2020 and 2030 will be uninterpretable in exactly the way Step 4’s 2010-to-2020 change already was, only more so.** The strongest answers connect this to Step 9 and notice that Wayne County contains Dearborn, home to the largest Arab-American concentration in the United States, whose residents the 2020 form counted as White — so the county the lab zooms into is one whose 2030 numbers will move for purely definitional reasons.

Watch for: a student assuming the 2024 standard is already in the data. It is not. It was issued in 2024, agencies’ action plans are now due in 2027 after two extensions, full compliance is due in 2029, and the Census Bureau has said it is *preparing* to implement it in the 2027 American Community Survey and the 2030 Census. Nothing in this lab’s data uses it.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given.
- **The move that earns a 4** is treating a definitional choice as a substantive decision with consequences, not as a technical footnote.
- **Also a 4:** quoting a number *with its universe attached* — “White alone,” “any-part Black,” “total population, not voting-age.” The documents model this everywhere; using it unprompted is the habit the lab exists to build.
- **Also a 4:** noticing that the random-shuffle null in Step 8 is what licenses the whole argument, and saying what the figure would look like without it.

- **A 2** is “race data is socially constructed so the numbers do not mean anything.” True in its first clause, false in its second, and the lab is built to move past it: the numbers mean something quite precise, and what they mean is a joint fact about people and about a directive.
- **Also a 2:** reporting the national composition as though it described anywhere.
- **Also a 2:** quoting Wayne County as a national finding, or the national staircase as a fact about Detroit. The documents make the scope explicit; ignoring it is the failure mode this lab introduces.
- **Reward anyone who asks what the numbers would be under the 2024 categories.** They cannot be computed, and knowing that is the right answer.

Anticipated questions

- “*Why is Hispanic not a race?*” — Because OMB said so in 1977 and again in 1997, and the Bureau implements OMB’s standard. As of the 2024 revision it is a category in the same list. The full answer is the **census** lab; the one-sentence answer is that this was an administrative decision with a date on it.
- “*Why Wayne County?*” — Because its county-level number is misleading in an instructive direction: no majority group, top 6.2% of counties for diversity, and most residents in a tract that is at least four-fifths one group. Then say what the national staircase is for: if the county were doing the work, Figure 6 would not be there.
- “*Is Wayne County typical?*” — On diversity, no: it is unusually high, which is the point. On the gap between its county score and its tract scores, it is a strong case rather than a unique one. The honest answer is that the lab does not compute the within-county gap for all 3,143 counties, and that doing so would be a good final project.
- “*Couldn’t you just use the dissimilarity index?*” — Yes, for two groups. The chapter says why it did not: with eight categories, choosing one contrast is a substantive claim. If a student wants Black-White dissimilarity for a particular metro, that is a legitimate extension, not a correction.
- “*Do the block numbers mean anything at all?*” — Individually, no. In aggregate, yes: the null in Step 8 shows most of the block-level effect survives. Say both halves.
- “*Has anything changed since 2020?*” — The categories have, on paper. The data has not. That gap — a standard in force and no data collected under it — is itself worth a minute.
- “*Why does the White population show up as both falling and rising?*” — Step 3 and Step 4 together. If a student can answer this question to the room, the session has worked.

Rebuilding the data

`data/build-data.R` downloads the 2020 national roll-up and all 51 state P.L. 94-171 files (about 1.3 GB, processed and discarded), the 2010 national file, and the TIGER layers not already cached elsewhere in this course. It reuses `../redlining/data/tiger/26/` for Michigan tracts and `../residual-votes/data/tiger/st/` for state outlines rather than re-downloading them. It writes about 1.7 MB of CSVs, records its source URLs and fetch date, and **stops** if the eight categories fail to partition the population at any level of any state, or if a state’s blocks do not sum to the state. Set `DEMOG_CACHE` to a directory of already-downloaded zips to skip the downloads. Rerun it and reknit all three documents; every number in them is computed at knit time, so a stale build is the only way they can go out of date.

Connections

- **census** — the two questions on the 2020 form, the write-in boxes, and “Some Other Race.” That lab explains where this lab’s Table P2 comes from. Run it first if you can; Step 1 assumes it.
- **areal-units** — the modifiable areal unit problem, on Fulton County. Step 8’s null is the control that lab’s ladder does not have, and the contrast is worth teaching explicitly.
- **census-geography** — the nesting spine. It is why Step 8’s ladder is assumption-free: a county’s population *is* the sum of its tracts, by construction, so nothing in Figure 6 required a crosswalk.

- **redlining** — one mechanism that could produce the step in Step 10, and the only lab in the course that puts a named document behind a boundary.
- **surnames and bisg-check** — what happens when race must be *inferred* because nobody recorded it. This lab is the case where somebody did.
- **section-203** — a federal formula that turns counts of specific language and ancestry groups into legal obligations, and therefore a place where the definitional choices in Steps 2 and 3 have direct consequences.